

# A Worked Numerical Example: Base Paper vs. Our Idea, Step by Step

We use **one running scenario** throughout – the same “hidden car behind a truck” situation the base paper itself uses as its motivating example – and plug in the base paper’s own numbers wherever possible ( $p_c = 0.1$ ,  $r_1 = 2$ ,  $r_3 = 50$ ,  $r_4 = 5$ ), so every comparison below is grounded in the real paper, not an abstract stand-in.

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## 1 The Shared Scenario

The AV is approaching a blind spot on the road (the paper’s own example: a car hidden behind a truck). Two things are uncertain at this moment:

1. **Is a hidden vehicle actually there?** Call this an “unexpected event.” We’ll use the same small probability the paper’s own setup implies for a risky moment:  $q = 0.05$  (a 5% chance an unexpected event is present).
2. **Is the communication channel currently good or bad?** The paper’s own number:  $P(c=1) = p_c = 0.1$  (10% chance the channel is bad).

The AV must choose radar mode ( $a=1$ , always detects the hidden vehicle, per the paper’s “perfect radar” assumption) or communication mode ( $a=0$ , sends data, but misses the vehicle entirely if it’s there).

## 2 Part 1 – Knowing the Channel State: Trusting a Single Reading vs. Building Belief

### 2.1 Setup

#### Numbers we'll use

- True prior chance the channel is bad:  $P(c=1) = 0.1$  (paper's own value).
- A real sensor reading (SINR estimate, ACK/NACK pattern, etc.) is right **80% of the time**:  $P(\hat{o} = c) = 0.8$ , wrong 20% of the time.
- Written as likelihoods:  $P(\hat{o}=1 \mid c=1) = 0.8$ ,  $P(\hat{o}=1 \mid c=0) = 0.2$  (and symmetrically for  $\hat{o} = 0$ ).

### 2.2 The base paper's way: trust the latest reading as fact

#### What the base paper effectively does

The base paper has no sensing step at all –  $c$  is simply handed to the AV as ground truth. The moment we try to actually deploy this on hardware, the natural (naive) way to bridge that gap is: take whatever the sensor says *right now*,  $\hat{o}_t$ , and feed it in as if it were  $c_t$  itself.

### 2.3 Our way: accumulate evidence with Bayes' rule

Instead, we track a **belief**  $b_t = P(c_t=1 \mid \text{all readings so far})$ , updated each time a new reading arrives:

$$b_t = \frac{P(\hat{o}_t \mid c=1) b_{t-1}}{P(\hat{o}_t \mid c=1) b_{t-1} + P(\hat{o}_t \mid c=0) (1 - b_{t-1})} \quad (1)$$

In words: *how much should this new reading shift what I already believed?* – not throw away everything I knew and trust only this one reading.

#### Step-by-step: three readings in a row say “bad”

Start:  $b_0 = 0.1$ .

**Reading 1,  $\hat{o}_1=1$  (bad):**

$$b_1 = \frac{0.8 \times 0.1}{0.8 \times 0.1 + 0.2 \times 0.9} = \frac{0.08}{0.26} \approx \mathbf{0.308}$$

**Reading 2,  $\hat{o}_2=1$  (bad again):**

$$b_2 = \frac{0.8 \times 0.308}{0.8 \times 0.308 + 0.2 \times 0.692} = \frac{0.246}{0.385} \approx \mathbf{0.640}$$

**Reading 3,  $\hat{o}_3=1$  (bad again):**

$$b_3 = \frac{0.8 \times 0.640}{0.8 \times 0.640 + 0.2 \times 0.360} = \frac{0.512}{0.584} \approx \mathbf{0.877}$$

So belief climbs  $0.1 \rightarrow 0.31 \rightarrow 0.64 \rightarrow 0.88$ : three consistent readings build real confidence that the channel is genuinely bad.

### Now a single contradicting reading arrives

**Reading 4,  $\hat{o}_4=0$  (suddenly says “good”):**

$$b_4 = \frac{0.2 \times 0.877}{0.2 \times 0.877 + 0.8 \times 0.123} = \frac{0.175}{0.274} \approx \mathbf{0.640}$$

**Naive approach (trust the latest reading only):** instantly flips to believing the channel is good ( $c=0$ ), full stop – discarding everything learned from the previous three readings.

**Our approach:** belief only drops from 0.88 to 0.64 – still leaning “probably bad.” One odd reading doesn’t erase three consistent ones.

### Why this is the improvement

If the channel really is bad and reading 4 was just a glitch, the naive approach gets fooled into picking communication mode at the worst possible moment. Our approach stays appropriately cautious because it weighs the *whole pattern* of evidence, not just whatever the sensor said one instant ago – exactly the robustness a real, noisy sensor needs and the base paper’s “perfect knowledge” assumption never had to worry about.

## 3 Part 2 – Guaranteeing Safety: A Hand-Picked Penalty vs. a Direct Target

### 3.1 Setup

Let  $\pi$  = the fraction of risky moments where the AV picks radar mode. Then the **miss-detection probability** is simply  $\mu = 1 - \pi$  (whenever it picks comm mode instead of radar, and an event is present, it’s a miss).

Using the paper’s own reward values –  $r_1 = 2$  (comm, no event),  $r_3 = 50$  (comm, event missed),  $r_4 = 5$  (radar, event caught) – and  $q = 0.05$ :

### Expected reward as a function of $\pi$

$$\mathbb{E}[\text{reward}](\pi) = \underbrace{(1 - \pi)[-qr_3 + (1 - q)r_1]}_{\text{comm-mode share}} + \underbrace{\pi[qr_4]}_{\text{radar-mode share}}$$

Plugging in numbers ( $q=0.05$ ,  $r_1=2$ ,  $r_3=50$ ,  $r_4=5$ ):

$$\mathbb{E}[\text{reward}](\pi) = (1 - \pi)(-0.6) + \pi(0.25) = -0.6 + 0.85\pi$$

This is a straight line in  $\pi$  with a *positive* slope (0.85), so the reward is maximized at the extreme  $\pi^* = 1$  (always use radar)  $\Rightarrow \mu^* = 0$ .

### The fragility problem: change the penalty a little, everything flips

Suppose instead of  $r_3 = 50$ , someone had picked  $r_3 = 10$  – still  $5\times$  larger than  $r_1$ , still looks like “a big safety penalty” at a glance.

$$\mathbb{E}[\text{reward}](\pi) = (1 - \pi)(-0.5 + 1.9) + \pi(0.25) = 1.4 - 1.15\pi$$

Now the slope is *negative*: the reward-maximizing choice flips to  $\pi^* = 0$  (**never** use radar)  $\Rightarrow \mu^* = 1$  – a **100% miss rate**. A  $5\times$  change in one hand-picked number swung the outcome from perfectly safe to completely unsafe, with nothing in between ever being reachable.

#### Why this isn't a one-off – it's structural

$\mathbb{E}[\text{reward}](\pi)$  is *always linear* in  $\pi$  for this kind of reward, regardless of which numbers are chosen:  $\mathbb{E}[\text{reward}](\pi) = A + \pi(B - A)$ . A straight line's maximum is always at an endpoint ( $\pi=0$  or  $\pi=1$ ) unless  $A = B$  exactly. So plain reward-shaping here can *never* land on a deliberately chosen middle ground, like “miss at most 5% of events” – it is mathematically stuck choosing between two extremes.

#### Our way: state the safety target directly, and solve for it

Instead of guessing a penalty and hoping, we pick the miss-rate we're willing to accept,  $\delta$ , up front – say  $\delta = 0.05$  – and directly require:

$$\mu = 1 - \pi \leq \delta \quad \Longrightarrow \quad \pi \geq 1 - \delta = 0.95$$

To get the most communication throughput while still respecting this, we use the *smallest*  $\pi$  that satisfies it:  $\pi^* = 0.95$  exactly  $\Rightarrow \mu^* = 0.05$  exactly – precisely the target we asked for, on purpose, every time – and unlike either reward-shaping outcome above, the AV still gets to use communication mode 5% of the time (nonzero throughput), instead of abandoning it completely ( $\pi=1$ ) or abandoning safety completely ( $\pi=0$ ).

## 4 Side-by-Side Summary

|                                           | Base paper (reward-shaping)                   | Our idea (direct targeting)                              |
|-------------------------------------------|-----------------------------------------------|----------------------------------------------------------|
| Channel belief after a noisy reading      | Whatever the latest reading says, full stop   | Weighted by everything observed so far                   |
| Resulting miss-rate with $r_3 = 50$       | $\mu = 0$ , but zero communication use at all | –                                                        |
| Resulting miss-rate with $r_3 = 10$       | $\mu = 1$ – complete safety failure           | –                                                        |
| Setting a target $\delta = 0.05$ directly | Not expressible – no $r_3$ reaches it         | $\mu = 0.05$ exactly, with real communication throughput |

**Takeaway:** the same numbers from the same paper, run through the same simple scenario, show two concrete things: (1) trusting a single noisy reading throws away useful history that a belief update would keep, and (2) a hand-picked reward weight can only ever push the AV to one of two extremes, never to a deliberately chosen safety level – while our constrained approach reaches that exact level, on purpose, every time.